A CASE STUDY ON RAW MATERIAL MANAGEMENT IN A MANUFACTURING INDUSTRY USING ADVANCED NONLINEAR OPTIMIZATION TECHNIQUE (NLOT):

BY

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Abstract

This study addresses raw-material inventory optimisation for Benmist Water Company, a water bottling establishment in Akure, Nigeria. The objective is to minimise the total cost of holding and ordering raw materials, given demand, holding cost, ordering cost, and storage-capacity parameters over a defined time period, with the order quantity as the decision variable.

Two optimisation algorithms are applied to this problem and compared: a Genetic Algorithm (GA), which iteratively refines a population of candidate order quantities through selection, crossover, and mutation, and Sequential Least Squares Quadratic Programming (SLSQP), a gradient-based method that solves the constrained problem directly. Both algorithms are run on the same problem instance, and their results are compared in terms of minimum cost achieved, number of iterations to convergence, and the shape of the underlying cost landscape.

The Genetic Algorithm converges to a minimum cost of 1085.0 at an order quantity of 7 units after 300 generations, while SLSQP converges to a minimum cost of 1083.6 at an order quantity of approximately 8 units in 8 iterations. The two algorithms agree closely on both the optimal order quantity and the minimum cost, with SLSQP reaching a comparable answer in substantially fewer iterations. This agreement, despite the two methods' very different search mechanisms, is itself a useful cross-check on the result. The comparison illustrates the practical trade-off between the Genetic Algorithm's broad, derivative-free exploration and SLSQP's fast, precise local convergence, and provides guidance for selecting an optimisation approach in similar single-item inventory problems.

1 Introduction

Efficient management of raw-material inventory is a central operational concern for manufacturing firms, and the water bottling industry is no exception. This report addresses the raw-material inventory optimisation problem faced by Benmist Water Company, a water bottling establishment in Akure, Nigeria. The objective is to minimise the total cost of holding and ordering raw materials, accounting for demand, holding costs, and ordering costs over a defined time period.

1.1 Background and Focus of the Study

The water bottling industry operates under dynamic demand and persistent cost pressure, which calls for a deliberate approach to inventory management [3]. Balancing the cost of holding excess inventory against the cost of placing frequent orders is a non-trivial optimisation problem [10]. Benmist Water Company's specific operating context motivates the optimisation approach developed in this study.

1.2 Problem Statement

Benmist Water Company must balance meeting customer demand, minimising holding costs, and controlling the costs associated with placing orders. This study addresses that problem using two optimisation algorithms, with the aim of providing actionable guidance for the company's inventory management practice.

1.3 Aim and Objectives

The aim of this study is to develop an order-quantity strategy for Benmist Water Company that minimises total inventory cost. The specific objectives are to:

1. minimise the total cost associated with holding and ordering raw materials;
2. determine the optimal order quantity using a Genetic Algorithm (GA); and
3. evaluate the effectiveness of Sequential Least Squares Quadratic Programming (SLSQP) for the same problem.

1.4 Study Question

How can a Genetic Algorithm and SLSQP be applied to minimise raw-material inventory costs for Benmist Water Company, given its specific demand pattern and operational constraints?

1.5 Scope and Limitations

This study is limited to the raw-material inventory management of Benmist Water Company. The optimisation model is fitted to this company's specific parameters and is not directly generalisable to other firms without re-estimation. The model assumes a steady demand pattern and considers only holding and ordering costs; external factors such as market trends and macroeconomic conditions are outside its scope.

2 Literature Review

2.1 Inventory Management in Bottled Water Manufacturing

Lean manufacturing principles call for the continuous reduction of waste across all business operations [1], which in a bottled water context requires close examination of production operations and the materials involved, including PET bottles, bottle caps, label materials, and catalyst additives [6]. Kumie [7] evaluated manufacturing waste and its effect on operational performance in a bottled water manufacturing company using a case study approach. Estimating an optimal order quantity requires characterising demand per period, holding cost per unit per period, ordering cost per order, initial inventory, storage capacity, and the planning horizon [12].

2.2 Optimisation Techniques for Raw-Material Inventory

A range of optimisation techniques has been applied to raw-material inventory problems in manufacturing. Oláh et al. [8] used inventory data exported from an SAP system to minimise inventory costs and maximise profit for a Hungarian production firm. Priymasiwi and Mustafid [9] applied a Differential Evolution algorithm to minimise raw-material inventory cost in a food-industry supply chain, based on material requirements, purchasing cost, safety stock, and reorder time. Thorat [11] optimised lot sizes for purchase parts and returnable transport items in a two-echelon closed-loop supply chain, applying SLSQP alongside Dual Annealing, Differential Evolution, and Simplicial Homology Global Optimisation.

Raw-material price volatility in rapidly changing markets makes purchasing strategy difficult for many manufacturers; Chen et al. [2] addressed this using a Genetic Algorithm to minimise total raw-material inventory cost. Ernawati et al. [4] used the Wagner-Whitin algorithm and the Silver-Meal heuristic to optimise order size and minimise inventory cost. Jainuri and Sukmono [5] applied a continuous-review model to determine the optimal number of orders for sodium caseinate inventory in the dairy industry. Together, these studies indicate that both evolutionary and gradient-based methods are established tools for this class of problem, which motivates comparing one representative of each family, GA and SLSQP, on the present case.

3 Problem Formulation

3.1 Decision Variable and Parameters

The decision variable is the order quantity of raw material, Q. The problem parameters are:

Symbol	Description
D	Demand for raw material (units per time period)
H	Holding cost per unit of raw material per time period
S	Ordering cost per order
I	Initial inventory level
C	Storage capacity (maximum inventory level allowed)
T	Total time period

3.2 Objective Function

The total cost to be minimised combines holding and ordering costs:

$$\min_{Q} \; Z(Q) = \frac{H}{2} \cdot \frac{Q}{2} + \frac{S \cdot D}{Q} + H \cdot I \tag{1}$$

3.3 Constraints

$$Q \geq D \cdot T \quad \text{(demand must be satisfied over the planning horizon)} \tag{2}$$

$$I + Q - D \leq C \quad \text{(inventory must not exceed storage capacity)} \tag{3}$$

$$Q, I \geq 0 \quad \text{(non-negativity)} \tag{4}$$

4 Methodology

Two algorithms from different optimisation families are applied to Equation 1 subject to Constraints 2–4: a Genetic Algorithm, which searches the solution space stochastically without requiring gradient information, and SLSQP, a deterministic gradient-based method. Running both on the same problem allows their solutions, convergence behaviour, and computational cost to be compared directly.

4.1 Genetic Algorithm

Genetic Algorithms are evolutionary optimisation techniques inspired by natural selection. Their appeal for inventory optimisation lies in their capacity to explore large solution spaces without requiring the objective function to be differentiable or convex, which suits the piecewise and constraint-penalised form the objective takes once feasibility penalties are added (see Section 4.3).

Figure 1 outlines the algorithm. Each generation evaluates the fitness of every candidate solution, applies crossover and mutation to produce the next generation, and repeats until a stopping criterion, here a fixed number of generations, is reached.

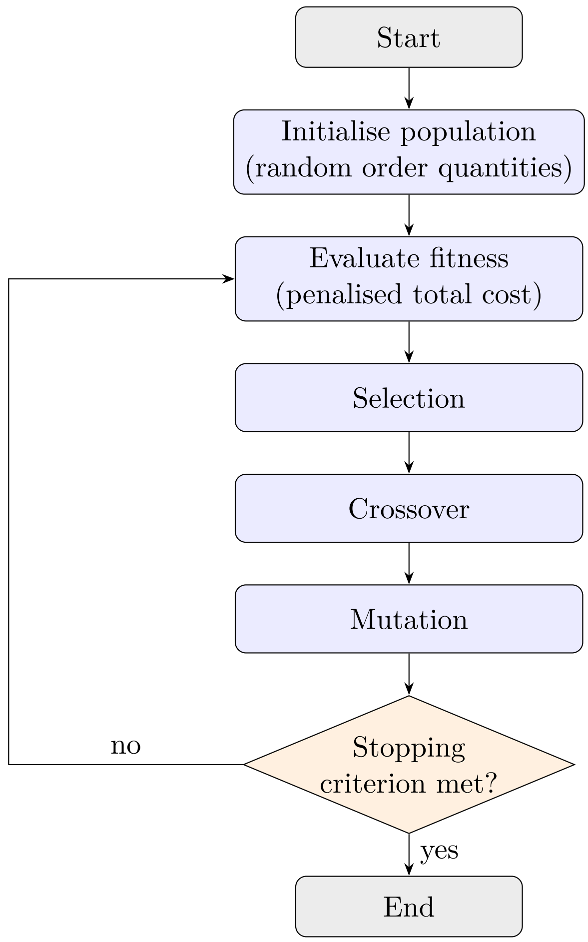

Figure 1: Genetic Algorithm flow.

The Genetic Algorithm is well suited to this problem because the objective is non-convex once constraint penalties are added, and its stochastic search reduces the risk of settling in a suboptimal region relative to a purely local search.

4.2 Sequential Least Squares Quadratic Programming (SLSQP)

SLSQP is a gradient-based method for constrained nonlinear optimisation. At each iteration, it solves a quadratic approximation of the problem to determine a search direction, then takes a step along that direction and re-evaluates the objective and constraints. Figure 2 summarises the procedure.

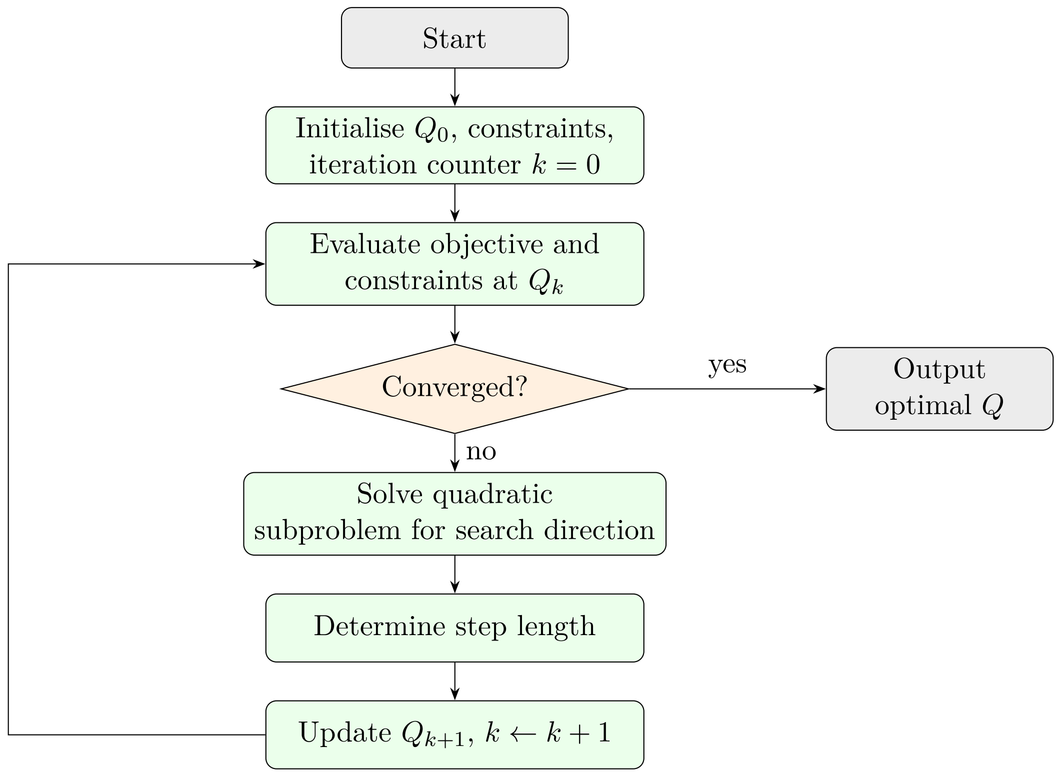

Figure 2: SLSQP flow.

SLSQP's deterministic, gradient-based convergence complements the Genetic Algorithm's stochastic exploration: where GA is well suited to broad exploration of the solution space, SLSQP is well suited to fast, precise refinement once a promising region has been located, which the parallel comparison in Section 5 makes explicit.

4.3 Implementation

Both algorithms are implemented in Python. The Genetic Algorithm uses the `pygad` library; constraint violations are handled by adding a large penalty to the cost whenever the demand or storage constraint is violated, since `pygad`'s fitness interface does not accept constraints directly. SLSQP uses `scipy.optimize.minimize` with `method="SLSQP"`, which accepts the inequality constraints directly. Full, documented source code is provided in the project repository rather than reproduced here; see the Appendix for the exact location of each script and how to reproduce every result and figure in this report.

5 Results and Discussion

5.1 Parameters Used

The case-study parameters, drawn from Benmist Water Company's operating data, are given in Table 1.

Table 1: Case-study parameters.

Parameter	Value
Demand, D	7 units per time period
Holding cost, H	20 per unit per time period
Ordering cost, S	50 per order
Initial inventory, I	50 units
Storage capacity, C	200 units
Time period, T	1 day

5.2 Optimisation Results

Table 2 summarises the outcome of each algorithm.

Table 2: Genetic Algorithm and SLSQP results.

Algorithm	Minimum cost	Iterations	Optimal order quantity
Genetic Algorithm	1085.0	300	7
SLSQP	1083.6	8	8

5.3 Genetic Algorithm Convergence

Figure 3 shows the Genetic Algorithm's objective value across 300 generations. The algorithm converges to a minimum cost of 1085.0 at an order quantity of 7 units. Figure 4 shows the corresponding order quantity trajectory: the population moves from a widely dispersed initial range toward the converged value within the first 150 generations, after which it remains stable.

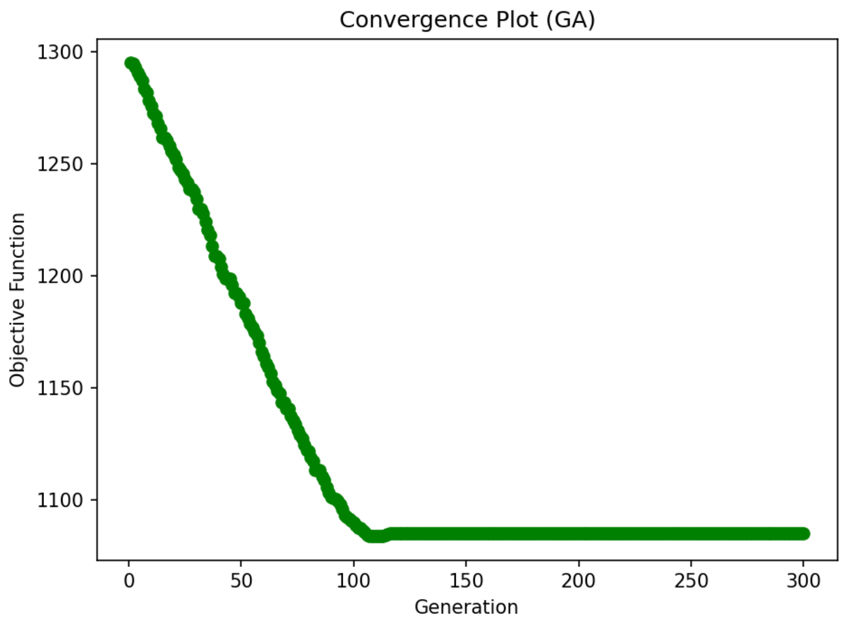

Figure 3: Convergence of the Genetic Algorithm's objective value across generations.

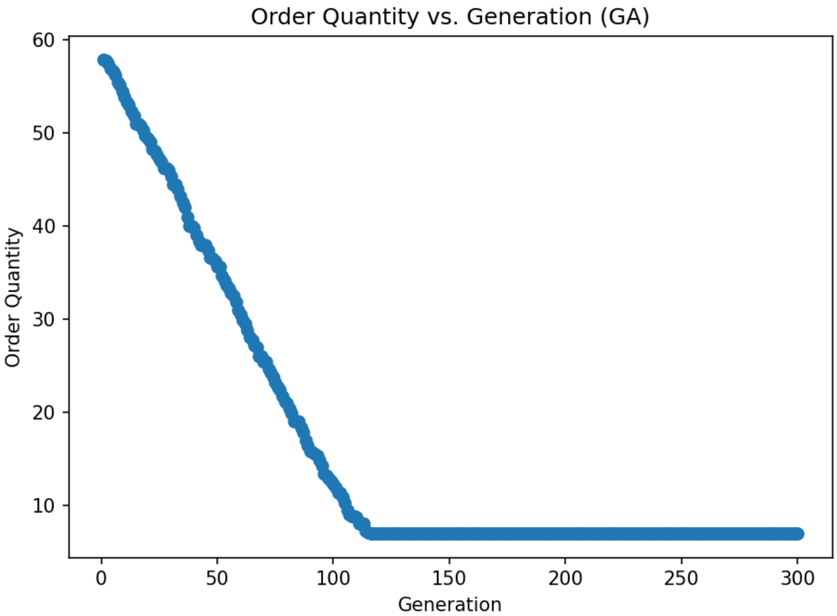

Figure 4: Order quantity of the best solution in each generation (Genetic Algorithm).

5.4 SLSQP Convergence

SLSQP reaches a minimum cost of 1083.6 in 8 iterations, with an optimal order quantity of approximately 8 units. Figure 5 shows the order-quantity trajectory across iterations; convergence is direct, without the broad initial search seen in the Genetic Algorithm, consistent with SLSQP's gradient-based, locally convergent design.

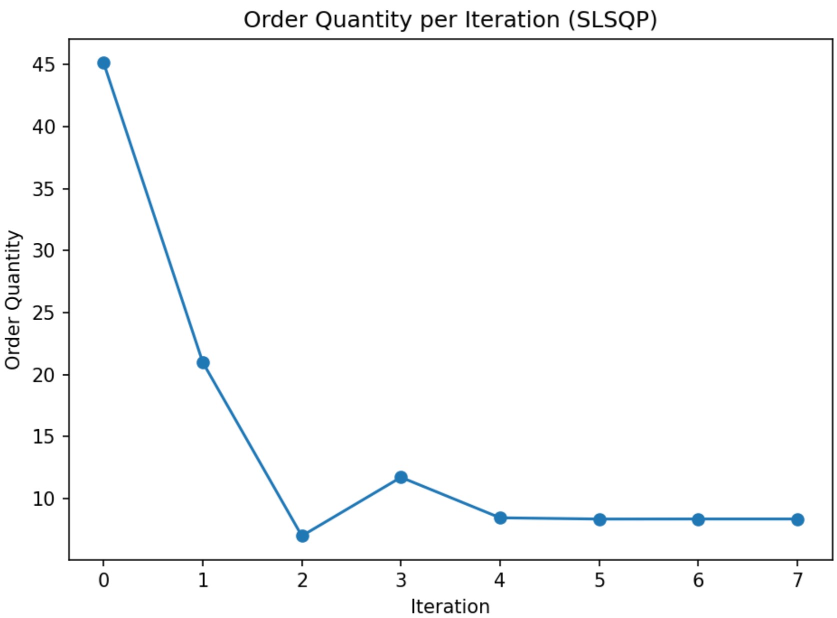

Figure 5: Order quantity at each SLSQP iteration.

5.5 Cost Landscape

Figure 6 plots total cost against order quantity directly from Equation 1. The curve has a single interior minimum, consistent with a classical economic-order-quantity trade-off: cost is dominated by the ordering term for small Q and by the holding term for large Q, with both algorithms' converged order quantities landing close to the minimum of this curve.

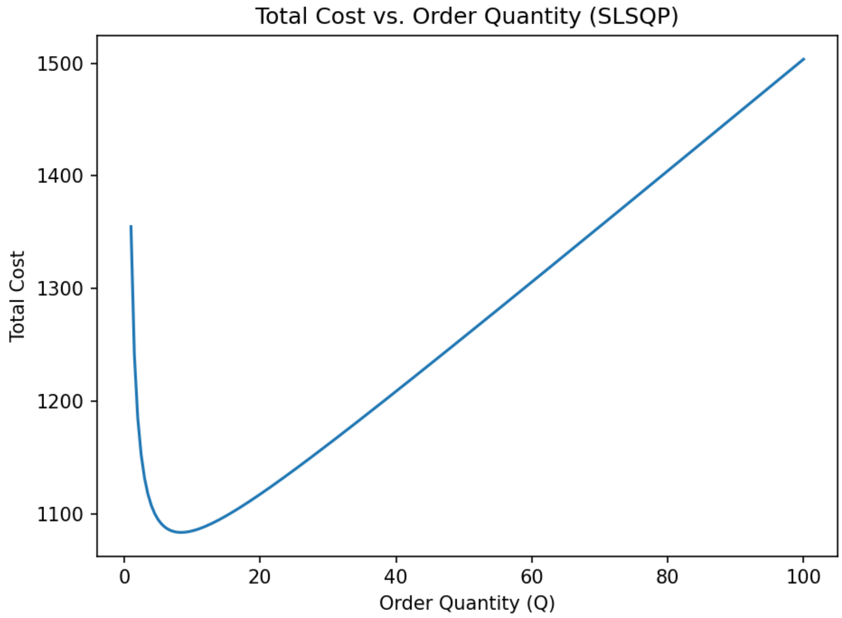

Figure 6: Total cost as a function of order quantity.

5.6 Comparison of the Two Algorithms

The Genetic Algorithm and SLSQP converge to closely agreeing answers: minimum costs of 1085.0 and 1083.6 respectively, a difference of about 0.1%, and order quantities of 7 and 8 units. This close agreement between a stochastic, derivative-free method and a deterministic, gradient-based one is a useful internal cross-check that neither algorithm has converged to a poor local optimum.

The two methods differ sharply in computational cost: SLSQP reaches its answer in 8 iterations, compared with 300 generations for the Genetic Algorithm. This gap reflects the structure of the problem rather than a general superiority of one method over the other; the objective in Equation 1 is smooth and has a single interior minimum once the constraint penalties are accounted for, which favours a gradient-based method such as SLSQP. The Genetic Algorithm's broader, population-based search is better suited to objective landscapes with multiple local minima or discontinuities, neither of which is present in this particular problem, which explains the difference in convergence speed observed here.

6 Conclusion

This study applied a Genetic Algorithm and SLSQP to the raw-material order-quantity problem faced by Benmist Water Company, minimising total holding-and-ordering cost subject to demand, storage, and non-negativity constraints. Both algorithms converge to closely agreeing solutions, a minimum cost of 1085.0 at an order quantity of 7 units for the Genetic Algorithm, and 1083.6 at approximately 8 units for SLSQP, with SLSQP reaching its solution in substantially fewer iterations. The agreement between the two methods supports the reliability of the result, and their contrasting convergence behaviour illustrates the practical trade-off between broad, derivative-free exploration and fast, gradient-based local refinement. For single-item inventory problems with a smooth, unimodal cost structure similar to the one studied here, these results suggest that a gradient-based method such as SLSQP offers a substantially faster route to the same answer, with a Genetic Algorithm remaining a useful cross-check or a preferred choice where the cost landscape is less well behaved.

Appendix: Code and Reproducibility

The Python implementation can be found in the project repository below:

`https://github.com/Olorunnisola01/raw-material-nlot`

- `src/ga_optimizer.py`: Genetic Algorithm implementation (Section 4.1).
- `src/slpsqp_optimizer.py`: SLSQP implementation (Section 4.2).
- `figures/`: every figure in this report, generated directly by the two scripts above.

References

[1] Becker, R. M. (1998). Lean manufacturing and the Toyota production system. *Encyclopedia of World Biography.*

[2] Chen, G.-H., Zhao, Y., & Su, B. (2017). Raw material inventory optimization for MTO enterprises under price fluctuations. *Journal of Discrete Mathematical Sciences and Cryptography*, 20, 255–270.

[3] Dunmade, I. S. (2017). Socioeconomic impacts of bottled water production and consumption system in a developing economy: A lifecycle approach. *International Journal of Business and Management Studies*, 6(02), 471–484.

[4] Ernawati, D., Dewi, S., Ni Ketut, S., & Budianto, K. (2021). Ordering size optimization of raw material to minimize inventory costs using Wagner-Whitin algorithm and Silver-Meal methods. *E3S Web of Conferences*, 328, 05002.

[5] Jainuri, V., & Sukmono, T. (2021). Optimization of inventory costs using the continuous review system (CRS) method in controlling the need for raw materials for the Crimean industry. *Academia Open*, 5.

[6] Kamoga Mulungi, S. M. (n.d.). *Water bottle manufacture and production.*

[7] Kumie, T. (2021). *Evaluation of manufacturing wastes and their impact on operational performances: The case of bottled water manufacturing company* [PhD thesis, St. Mary's University].

[8] Oláh, J., Lakner, Z., Hollósi, D., & Popp, J. (2017). Inventory methods in order to minimize raw materials at the inventory level in the supply chain. *LogForum*, 13(4), 439–454.

[9] Priymasiwi, M., & Mustafid. (2018). Optimization of raw material inventory costs in the food supply chain using Differential Evolution algorithm. *E3S Web of Conferences*, 73, 13016.

[10] Pulido-Rojano, A., Pizarro-Rada, A., Padilla-Polanco, M., Sánchez-Jiménez, M., & De-la-Rosa, L. (2020). An optimization approach for inventory costs in probabilistic inventory models: A case study. *Ingeniare. Revista Chilena de Ingeniería*, 28(3), 383–395.

[11] Thorat, P. H. (2020). *Lot size optimization with returnable transport item in closed-loop supply chain* [PhD thesis, State University of New York at Binghamton].

[12] Zheng, X., & Chen, Y. (2024). Optimization of inventory cost control for SMEs in supply chain transformation: A case study and discussion. *E+M Ekonomie a Management*, 1–21.